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Paradigms of Machine Learning - I

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Types of Machine Learning: A Quick Recap

Supervised Learning

Key Components of Text-to-speech Systems

- Uses labelled input-output pairs
- Learns to map inputs to outputs

Types:

1

Classification: Predicts categories

Example: Email spam detection (spam / not spam)

2

Regression: Predicts continuous values

Example: House price prediction

Unsupervised Learning

Finding Patterns Without Labels

Pattern Discovery

Learning from data without labels to find hidden patterns

Key Algorithms

K-means, DBSCAN, PCA, autoencoders

Real Uses

Customer segmentation, feature extraction

Semi-supervised Learning

Learns from a Mix of Labelled and Unlabelled Data

Semi-Supervised Learning combines small labelled data with large unlabelled data for improved training

- Uses few labelled and many unlabelled examples
- Cuts labelling cost for large datasets
- Learns patterns from unlabelled data

Semi-supervised Learning

Learns from a Mix of Labelled and Unlabelled Data

Example: Customer Reviews (Positive/Negative)

- 1,000 labelled + 9,000 unlabelled reviews
- Train with labelled data
- Predict labels for unlabelled data and retrain

Self-Supervised Learning

Learning by Creating Labels from Unlabelled Data

Process

- **Generate Signals**
  Extract supervision from unlabelled data
- **Create Pretext Tasks**
  Design tasks that produce automatic labels
- **Pre-train the Model**
  Use tasks like masked language prediction
- **Transfer Knowledge**
  Apply to real tasks with far less labelled data

Active Learning

Smart Labelling – The Model Chooses What to Label

Active Learning lets the model select data points for labelling to optimise learning while reducing labeling costs

Active Learning Workflow:

- Best when labelling is costly or slow
- Train on small labelled data
- Select and label uncertain samples
- Retrain and repeat

Example – Medical Imaging:

Task: Classify X-ray or MRI scans as normal/abnormal

Why? Radiologist time is costly

Use: Labels only unclear scans

Transfer Learning

A Quick Overview

Pre-train on Source: Learn general features from large datasets

Transfer Knowledge: Adapt features to new areas

Fine-tune on Target: Optimise for specific tasks with minimal data

Transfer learning reduces training time by 75% and data needs by 90%

Zero-shot Learning

Definition: The model predicts unseen classes without labeled training examples

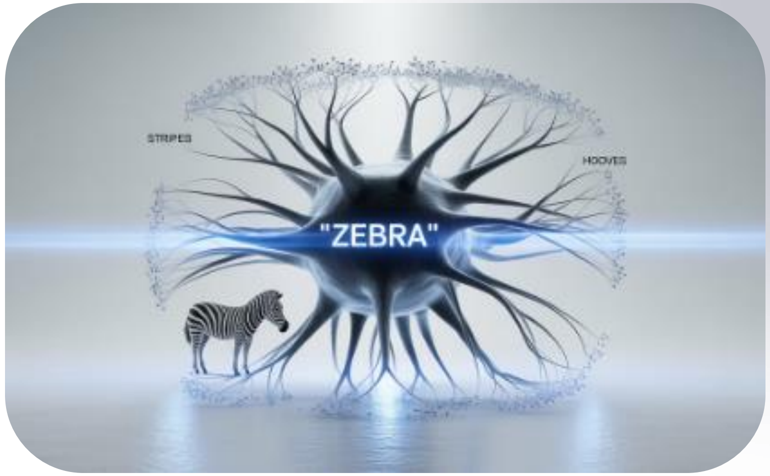
Key Idea: Relates unseen and seen classes using semantic information

Example: A model trained on cats, dogs, and birds identifies a zebra using the description "an animal with black and white stripes"

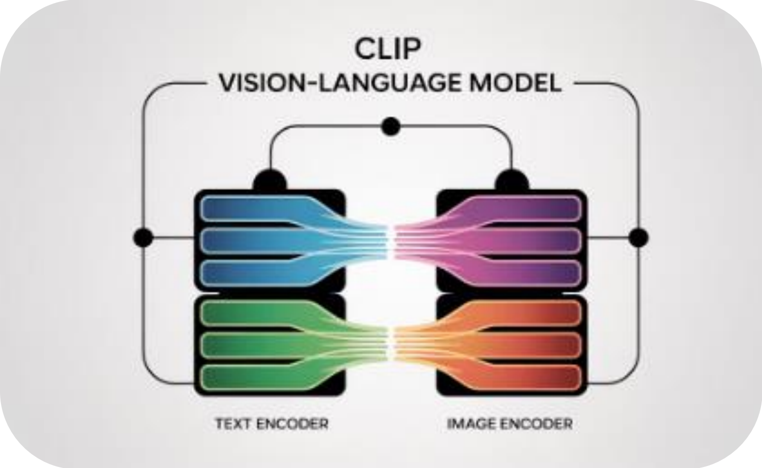
Used in: NLP tasks (e.g., sentiment or intent classification) and vision-language models like CLIP

Zero-shot Learning

Zero-shot Recognition: Identifies new classes using only semantic descriptions



Vision-Language Models: Use text-image links, like in CLIP



One-shot vs Few-shot Learning

Learning from Minimal Examples

One-shot Learning

- Learns from a single labelled example per class
- Based on similarity or meta-learning
- **Example**: Recognises a face from just one photo
- Used in Face ID, signature checks, image recognition

Few-shot Learning

- Learns from a few labelled examples (2–100)
- Adapts quickly with little data
- **Example**: Chatbot learns from 5 sample queries per issue type
- Used in chatbots, intent detection, language translation

Warm-up Activity

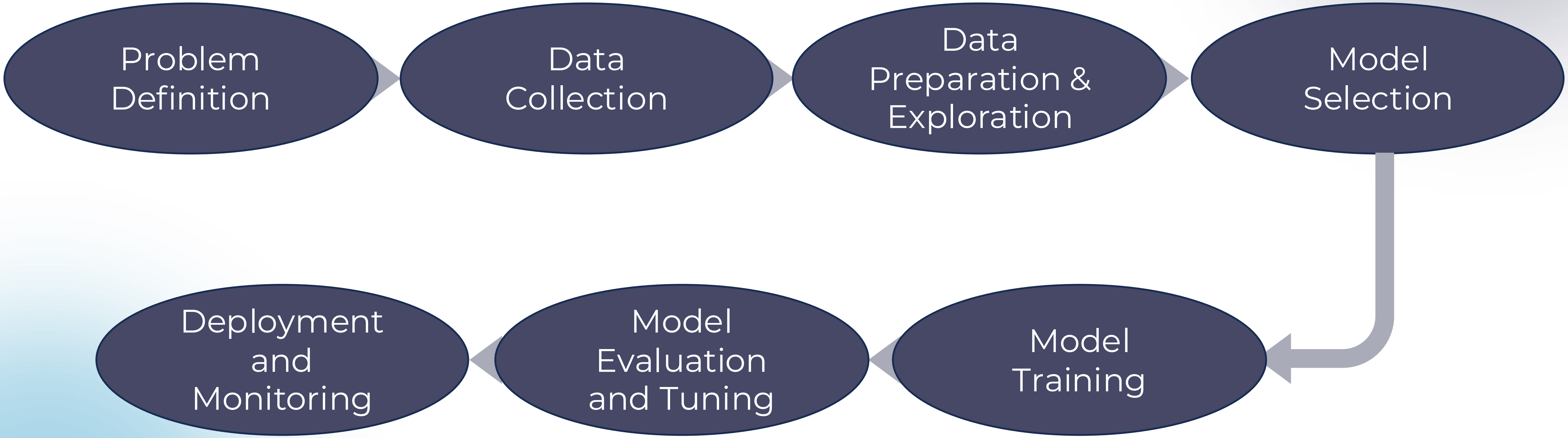
Match Each Scenario to the Correct Learning Type

Example	Your Answer (Learning Type)
1. Fraud Detection	
2. Topic Clustering on News Articles	
3. Medical Image Labeling with Limited Data	
4. Personalised Voice Assistant	
5. ChatGPT Zero-shot Capabilities	

Machine Learning Workflow

Machine Learning Workflow

Key Steps in Building ML Solutions



Step 1: Problem Definition

Set Clear Goals and Approach

Business Objective:
Clearly state the problem to solve (e.g. fraud, image recognition, churn)

Approach Selection:
Choose between supervised or unsupervised learning

Success Metrics:
Set how success will be measured (e.g. accuracy, speed, value)

Step 2: Data Collection

Gather Relevant and Reliable Data

- Use internal sources (e.g. logs, databases)
- Access external APIs (e.g. market or social media data)
- Explore open datasets (e.g. ImageNet, gov data)
- Check data quality (completeness, accuracy, relevance)

Step 3: Data Preparation and Exploration

Clean, Understand and Get Data Ready

Clean

Fix nulls, errors, outliers

Explore

Analyse patterns, compute stats and understand relationships

Engineer

Create and transform features

Split

Separate train and test sets

Step 4: Model Selection

Key Models and Their Trade-Offs

Neural Networks

- Best for: Complex patterns, image/speech recognition
- Trade-off: Needs large data, high computation

Decision Trees

- Best for: Intuitive models, classification tasks
- Trade-off: Can overfit without tuning

Linear Models

- Best for: Simple relationships, small datasets
- Trade-off: Misses complex patterns

Support Vector Machines

- Best for: Clear-boundary classification
- Trade-off: Sensitive to parameters

Step 5: Model Training

Key Steps for Effective Training

Data Splitting

- Training set: 80%
- Test set: 20%
- Keep validation data separate

Algorithm Application

- Feed data to chosen model
- Address class imbalance

Validation Techniques

- Use k-fold cross-validation
- Test across data subsets

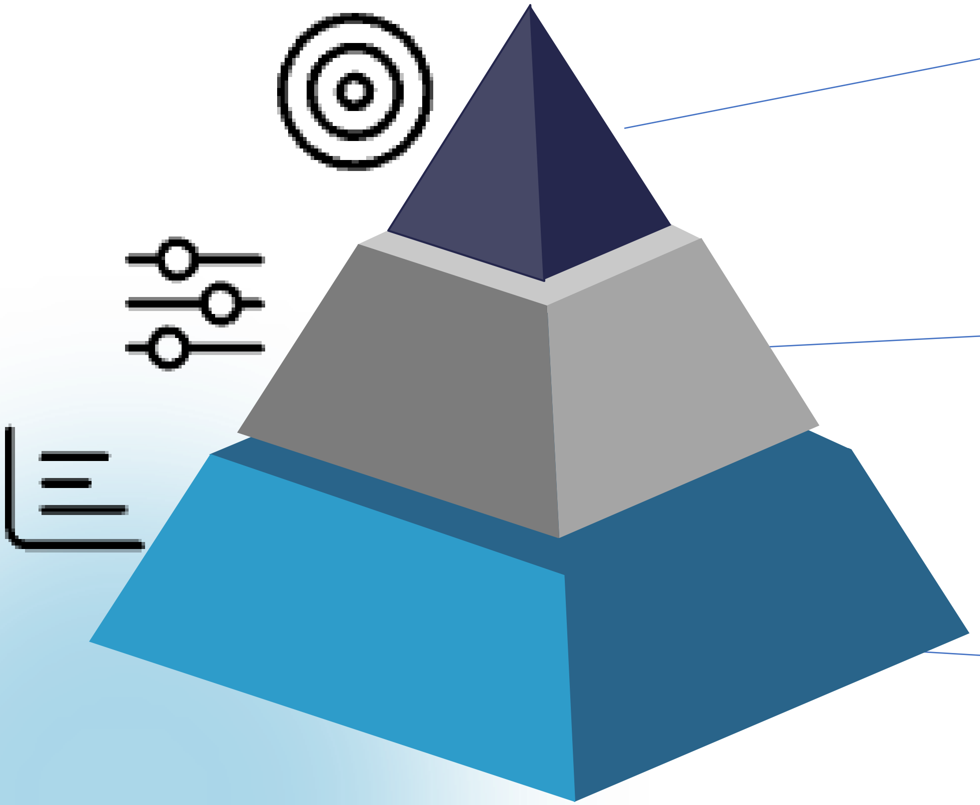
Initial Assessment

Check for:

- Underfitting
- Overfitting
- Convergence issues

Step 6: Model Evaluation and Tuning

Key Metrics and Optimisation Techniques



Performance Improvement

F1 score: 0.64 → 0.82

Hyperparameter Tuning Methods

Grid search Random search Bayesian optimisation

Key Evaluation Metrics

Classification: Precision, Recall, F1, ROC-AUC

Regression: Mean squared error

Step 7: Deployment and Monitoring

Ensuring Model Reliability

Service Standards

- 99.9% uptime target
- 24/7 performance monitoring

Model Maintenance

- Regular retraining (e.g., 30-day cycles)
- Adapt to real-world data drift

Deployment Options

- Cloud APIs
- Web applications
- Embedded systems

Key Focus

- Continuous performance tracking
- Maintain user experience quality

Introduction to Machine Learning Tools

Why Use Machine Learning Tools?

Key Benefits for Your Workflow

- Automate repetitive tasks
- Boost accuracy and productivity
- Support team collaboration
- Power end-to-end ML workflows

Problem Definition

Aligning ML With Business Goals

Core Objective

- Understand business needs
- Define ML problem statement

Key Tools

- Process mapping: Miro, Lucidchart
- Documentation: Notion, Trello
- Domain data: CRMs, ERP systems

Data Collection

Gathering Raw Data from Multiple Sources

Key Objective

- Acquire and consolidate raw datasets

Essential Tools

- Streaming: Apache Kafka, NiFi
- Web scraping: BeautifulSoup, Scrapy
- Structured data: APIs, SQL, BigQuery
- Data handling: Python Pandas

Data Preparation and Exploration

Tools and Tasks

Data Manipulation
pandas, NumPy, Dask

Interactive Wrangling
OpenRefine, Trifacta

Visualisation and Exploration
Matplotlib, Seaborn, Plotly

Goal: Clean, transform and understand data

Model Selection

Algorithms and Tools

Classical ML
Scikit-learn

AutoML and Interpretability
H2O.ai

Gradient Boosting
XGBoost, LightGBM

Goal: Select the optimal model

Model Training

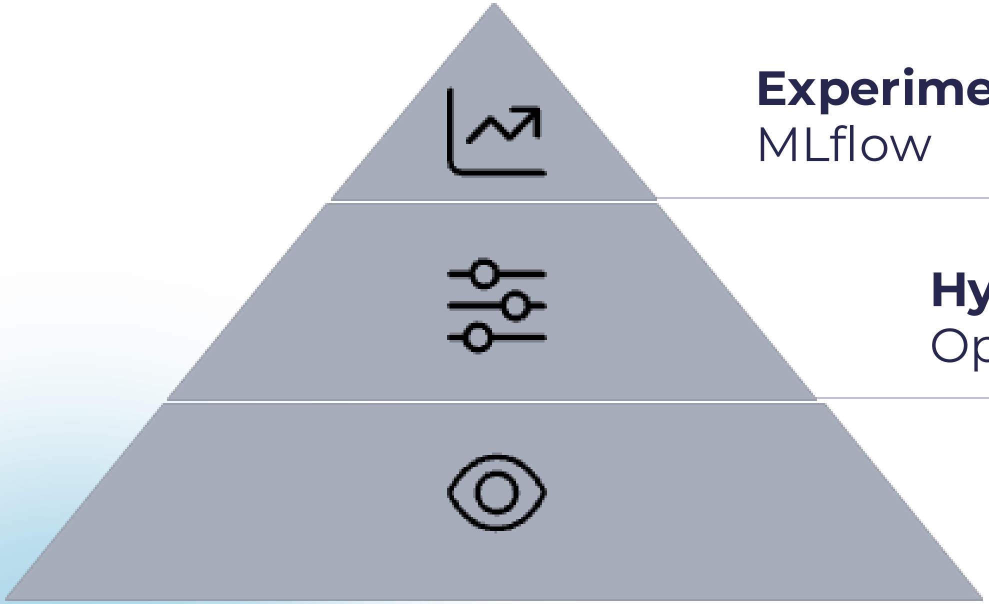
Frameworks and Tools

- **Deep Learning**
  TensorFlow, PyTorch
- **High-level API**
  Keras
- **GPU Acceleration**
  RAPIDS

Goal: Train model efficiently

Model Evaluation and Tuning

Key Components



Experiment Tracking
MLflow

Hyperparameter Tuning
Optuna, Hyperopt

Visualisation and Monitoring
Weights and Biases, Comet.ml

Goal: Optimise model performance

Deployment and Monitoring

Key Components

Model Serving

Docker, FastAPI, Flask

Performance Tracking
Prometheus, Grafana, Evidently AI

Scalable Infrastructure
Kubernetes, AWS SageMaker, Azure ML

Goal: Maintain reliable model performance in production

Demo – ML Lifecycle with Deployment and Monitoring

ML Lifecycle Demo

End-to-End Workflow

Objective:
Build and deploy a breast cancer classifier using Python

Stage	Activity	Tool
Problem Definition	Classify breast cancer as benign/malignant	Built-in use case
Data Collection	Load Breast Cancer dataset from scikit-learn	sklearn.datasets
Data Preparation and Exploration	Split, scale, inspect features and labels	pandas, scikit-learn
Model Selection and Training	Random Forest Classifier	sklearn.ensemble
Evaluation	Classification report with precision, recall, f1-score	sklearn.metrics
Deployment	Interactive prediction app	gradio
Monitoring	Classification performance dashboard	evidently

Q&A Session

- Open floor for questions and clarifications.